Fiscal Year:	FY 2026	**Task Last Updated:**	FY 01/08/2026
PI Name:	Bershad, Eric M. M.D.		
Project Title:	The Long-Term Consequences of Spaceflight on Brain and Eye Health		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2033 per NSSC information (Ed., 6/15/23) NOTE: End date changed to 3/10/2023 per NSSC information (Ed., 3/18/22)		
Key Personnel Changes/Previous PI:	Per the PI: Dr. Sangi-Haghpeykar left the institution (Baylor College of Medicine) so she is no longer serving as a Co-Investigator (Ed., 1/21/26).		
COI Name (Institution):	Hasan, Khader Ph.D. (University of Texas Health Science Center at Houston) Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Hirzallah, Mohammad M.D. (Baylor College of Medicine, Inc.) Ivkovic, Vladimir Ph.D. (Massachusetts General Hospital) Strangman, Gary Ph.D. (Massachusetts General Hospital) Strutt, Adriana Ph.D. (Baylor College of Medicine, Inc.) Chethan, Venkatasubba Rao M.D. (Baylor College of Medicine, Inc.) York, Michele Ph.D. (Baylor College of Medicine, Inc.)		
Grant/Contract No.:	80NSSC21K0757		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The key objectives of this study include: 1. Determination of the long-term health consequences of Spaceflight Associated Neuro-ocular Syndrome (SANS) and related ocular changes from long-duration spaceflight; 2. Assess the long-term health consequences of SANS-related brain alterations. Background: Long-duration spaceflight causes structural changes in the brain and eye, including the development of the Spaceflight Associated Neuro-ocular Syndrome (SANS) in about one-third of astronauts. Furthermore, cerebral venous outflow abnormalities including stasis, reversal of flow and/or venous thrombosis in the internal jugular vein (IJV) were recently observed in some astronauts. Given that the IJVs are the primary cerebral venous outflow pathway in the supine position on Earth, and both ocular venous and cerebrospinal fluid (CSF) outflow are dependent on the IJVs, a common pathway that may explain abnormalities in both the brain and eye following long-duration spaceflight may relate to changes in the venous outflow of the brain and eye due to microgravity. While the etiology of SANS and other brain and/or ocular structural changes are starting to be elucidated, knowledge about the long-term consequences to astronaut health remains limited. The focus of our study will be to assess long-term consequences of SANS “exposure,” as measured by three key anatomical/functional changes in astronauts that may relate to eye and/or brain health: (1) total retinal thickness as an assessment of optic disc edema, (2) the internal jugular vein (IJV) blood flow status, and (3) brain structure and compartment volumes. Aim 1: Assess the long-term health consequences of SANS-related ocular alterations. We will first consolidate retrospectively collected data relevant to health outcomes of interest of the eye and brain. Retrospective data will include optical coherence tomography (OCT)–derived total retinal thickness and choroidal thickness, magnetic resonance imaging (MRI)-derived brain and eye volumetric data, and vascular imaging of the brain and neck vessels including arteries and veins, as available from existing sources (Lifetime Surveillance of Astronaut Health (LSAH)/Life Sciences Data Archive (LSDA)). Prospectively, we will collect pre- and post-flight data for up to 5 years in astronauts on the following set of health outcome measures: (1) functional visual measures (visual acuity, visual fields, electroretinography), (2) Neurological examination and neuropsychological cognitive testing, (3) biomarkers of cognitive decline (ApoE4, A#42#A#40, and neurofilament light), and (4) MRI brain derived structural and volumetric measurements. These domains are of inherent health concern and focus on relevant functional outcomes that one might predict to be affected by spaceflight-induced changes in brain and ocular performance. Hyp1.1: Retinal thickening pre- vs. post-flight will predict long-term ocular functional change. Aim 2: Assess the long-term health consequences of SANS-related brain alterations. Using the same database as generated for Aim 1, we will conduct a parallel investigation of the long-term effects (up to 5 years) of brain-related changes in long-duration astronauts. Hyp2.1: IJV flow abnormalities during long-duration spaceflight will predict long-term cognitive changes and MRI volumetric changes in cerebrospinal fluid volume. Hyp2.2: Brain structural changes pre- vs. post-flight will predict long-term balance and cognitive or operationally relevant performance change. Significance: The proposal will provide critical information regarding long-term health consequences of spaceflight on the brain and eye in astronauts. In addition, it will help identify and quantify the extent to which any long-term health consequences can be predicted by the currently most-relevant markers of SANS. Ed. Note (2/29/24): This project has been combined with the work of three other Principal Investigators (PIs) who responded to the same solicitation with independent proposals. The protocols and aims from all four PIs have been integrated into a revised proposal that was delivered to NASA Human Health Countermeasures (HHC) and Human Factors and Behavioral Performance (HFBP) Element Scientists in January 2022. For information on the four related investigations, see: • The Long-Term Consequences of Spaceflight on Brain and Eye Health (PI: Bershad). • Long-Term Brain Structural and Functional Consequences of Spaceflight (PI: Basner). • Investigating Long-term Structural and Functional Changes in the Eye and Brain After Spaceflight (PI: Macias). • Recovery Timeline of Spaceflight-Induced Central Nervous System Changes (PI: Seidler). Ed. Note (01/2025). In 2025, due to logistical issues related to the ISS operations, the longitudinal aims of the study were halted. Instead, astronauts (n=20) who previously flew on a long-duration mission of at least 6 months and > 5 years prior will be called back for a comprehensive neurological and ocular assessment. Age and sex matched controls will be recruited for comparison.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide the longest and most comprehensive long-term follow up data of astronauts brain and eye health after a long-duration mission. This will help inform scientists, physicians, and the general public on the long-term risks of spaceflight. The study includes a comprehensive battery of neuroimaging, cognitive tasks, neurological exam, visual and ocular data, sensorimotor performance that will give a well developed structural and functional evaluation of brain and eye health.

Task Progress:	Data collection was initiated in August 2025 and has occurred in two crewmembers who previously flew on long-duration spaceflight missions. Ocular testing included optical coherence tomography, electroretinography, intraocular pressure, visual field analysis, optical biometry, and orbital magnetic resonance imaging (MRI). Multi-modal brain MRI will assess resting state functional MRI, brain volume, lateral ventricle size, pituitary gland deformation, and glymphatic function. Comprehensive neurological examination and neuropsychology testing was performed to correlate MRI findings with cognitive and neurological functioning. Blood samples were obtained to identify key neurotrophins and neurodegenerative biomarkers. One of these crewmembers has completed all testing procedures and the second crewmember has completed half of the testing procedures. Recruitment of control participants was initiated in October 2025.
Bibliography Type:	Description: (Last Updated: 06/17/2026)